LAB#12 MACROS

Task#01:

Write a program that takes single character input from the user and display in new line

```
NEW_LINE     MACRO    LN, CR   ;SET CURSOR
             MOV AH, 02
             MOV DL, LN
             INT 21H
             MOV DL, CR
             INT 21H
             ENDM

DISPLAY MACRO    STRING       ;DISPLAY MESSAGE
             MOV AH, 09
             MOV DX, OFFSET STRING
             INT 21H
             ENDM


             .MODEL  SMALL
             .STACK  64H

             .DATA
      MSG1   DB  'Enter A Character: $'
      MSG2   DB  'The typed character is: $'

             .CODE
      MAIN   PROC     FAR
             MOV      AX, @DATA
             MOV      DS, AX

             DISPLAY MSG1

             MOV AH, 1
             INT 21H
             MOV BH, AL

             NEW_LINE 0AH, 0DH

             DISPLAY MSG2

             MOV AH, 2
             MOV DL, BH
             INT 21H

             MOV AH, 4CH
             INT 21H
      MAIN   ENDP
             END MAIN
```

OUTPUT:

```
emulator screen (80x25 chars)
Enter A Character: M
The typed character is: M
```

TASK#02:

Write a program to display your bio data using carriage return and linefeed macro

```
NEW_LINE    MACRO   LN, CR
            MOV AH, 02
            MOV DL, LN
            INT 21H
            MOV DL, CR
            INT 21H
            ENDM

DISPLAY     MACRO   STRING
            MOV AH, 09
            MOV DX, OFFSET STRING
            INT 21H
            ENDM

INPUT       MACRO   BUFF
            MOV AH, 0AH
            MOV DX, OFFSET BUFF
            INT 21H
            ENDM



        .MODEL  SMALL
        .STACK  64H

        .DATA
 MSG    DB  '-:BIO-DATA:-$'
 MSG1   DB  'NAME         : $'
 MSG2   DB  'FATHER NAME  : $'
 MSG3   DB  'QUALIFICATION: $'
 MSG4   DB  'CGPA         : $'
 MSG5   DB  'DATE OF BIRTH: $'
 MSG6   DB  'CONTACT NO.  : $'

 IN1    DB  20,?,20 DUP(?)


        .CODE
 MAIN   PROC    FAR
        MOV     AX, @DATA
        MOV     DS, AX

        DISPLAY MSG
        NEW_LINE 0AH, 0DH
        NEW_LINE 0AH, 0DH

        DISPLAY MSG1
        INPUT IN1
        NEW_LINE 0AH, 0DH

        DISPLAY MSG2
        INPUT IN1
        NEW_LINE 0AH, 0DH

        DISPLAY MSG3
        INPUT IN1
        NEW_LINE 0AH, 0DH

        DISPLAY MSG4
        INPUT IN1
        NEW_LINE 0AH, 0DH

        DISPLAY MSG5
        INPUT IN1
        NEW_LINE 0AH, 0DH

        DISPLAY MSG6
        INPUT IN1
        NEW_LINE 0AH, 0DH


        MOV AH, 4CH
        INT 21H
 MAIN   ENDP
        END     MAIN
```

OUTPUT:

emulator screen (80x25 chars)

```
-:BIO-DATA:-

NAME          : MOHIB UDDIN
FATHER NAME  : SAIF UDDIN
QUALIFICATION: BSCS
CGPA          : 3.98
DATE OF BIRTH: 31 OCT 2021
CONTACT NO.  : 03122061060
```

TASK#03:

Write a macro that multiplies two words by repeated addition, and then saves the result. Write a program that multiplies the following numbers using macro:

- 2000 x 500
- 2500 x 500
- 300 x 400

```
MULTIPLY      MACRO VAL1, VAL2, RES
              LOCAL BACK

              MOV BX, VAL1
              MOV CX, VAL2
              MOV AX, 0
              MOV DX, AX
BACK:         ADD AX, BX
              ADC DX, 0
              LOOP BACK
              MOV RES, AX
              MOV RES+2, DX
              ENDM

              .MODEL SMALL
              .STACK 64H

              .DATA
              RES1     DD ?
              RES2     DW  2 DUP(?)
              RES3     DW  2 DUP(?)

              .CODE
MAIN      PROC
          MOV AX, @DATA
          MOV DS, AX

          MULTIPLY 2000, 500, RES1
          MULTIPLY 2500, 500, RES2
          MULTIPLY 300, 400, RES3

          MOV AH, 4CH
          INT 21H

MAIN      ENDP
          END MAIN
```

TASK#04:

Write a macro to calculate total wages which is equal to the sum of overtime, salary and bonuses. Write a program by using following values:

- Salary = 60000, Overtime = 2500 and Bonuses = 3000
- Salary = 45000, Overtime = 1000 and Bonuses = 500
- Salary = 10000, Overtime = 500 and Bonuses = 100

```
WAGES   MACRO SALARY, OVERTIME, BONUS
        MOV AX, 0
        MOV DX, AX

        ADD AX, SALARY
        ADD AX, OVERTIME
        ADC DX, 0
        ADD AX, BONUS
        ADC DX, 0
        ENDM

        .MODEL SMALL
        .STACK 64H

        .DATA

        .CODE
MAIN    PROC FAR
        MOV AX, @DATA
        MOV DS, AX

        WAGES 60000, 2500, 3000
        WAGES 45000, 1000, 500
        WAGES 10000, 500, 100

        MOV AH, 4CH
        INT 21H

MAIN    ENDP
        END MAIN
```